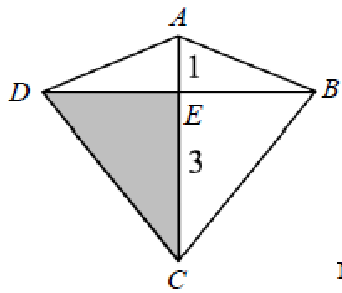
1	**YTSS/2025/I/01**
	Given that p is 30% of q, find the ratio of $4p : 3q$, giving your answer to its simplest form [2]
2	**YTSS/2025/I/15**
	A park is drawn on a map with a scale of 1 : 80 000. **(a)** The distance from the carpark to the playground is 1.5 cm on the map. Find the actual distance, in kilometres, from the carpark to the playground. [1] **(c)** The actual area of the park is 1.85 km^2. Find the area of the park on the map in square centimetres. [2]
3	**YTSS/2025/I/12**
	The formula for **Pendulum Period Formula (Simple Pendulum)** is given by $T = 2\pi\sqrt{\frac{L}{g}}$, where T: Period of the pendulum (time for one full swing) L: Length of the pendulum g: Gravitational acceleration (the value of g remains the same at the same location) An experiment is conducted at a fixed location to investigate the relationship between T and L. **(a)** Assuming that air resistance is negligible, using the formula, explain why T is directly proportional to square root of L. [1] **(b)** A change in L produces a change in T. When L is increased by 300%, find the percentage change in the value of T. [2]
4	**SCGS/2025/I/21**
	The time, t minutes, taken to fill up a tank completely is inversely proportional to the number of taps, n, used. A tank is filled up completely with 7 taps in 40 minutes. **(a)** If an additional tap is used, find the time taken for the tank to be filled up completely. [2] **(b)** Explain why 3 additional taps are needed for the tank to be filled up completely 12 minutes faster. [2]
5	**SST/2025/I/05**
	It is given that y is directly proportional to x^2. Find the percentage increase in y if x is increased by 25% of its original value. [2]

6	**SST/2025/I/11**
	The diagram shows a kite $ABCD$. AC and AB are the diagonals of the kite. E is a point on AC such that $AE : EC = 1 : 3$.  Not drawn to scale Find the fraction of the kite that is shaded. [2]
7	**GMSS/2025/I/21**
	y is inversely proportional to the cube root of $(3x + 2)$ and $y = 3$ when $x = 2$. Find y in terms of x. [3]
8	**ACSBR/2025/I/07**
	Given that y is inversely proportional to the cube root of x and $y = 50$ for a particular value of x. Find the new value of y when the value of x is multiplied by 8. [2]
9	**PHS/2025/I/06**
	Given that $A : B = \frac{1}{6} : \frac{1}{5}$ and $B : C = 3 : 7$, find $A : C$ as a ratio of two integers. [2]
10	**PHS/2025/I/08**
	The force F, between two particles is inversely proportional to the square of the distance between them. The force is 64 units when the distance between the particles is r metres. Find the force when the distance between the particles is reduced by 20%. [2]

Solution

1	$p=\frac{30}{100}q---[\text{M1}]$ $\frac{p}{q}=\frac{3}{10}$ $\frac{4p}{3q}=\frac{4\times3}{3\times10}$ $\frac{4p}{3q}=\frac{2}{5}$ $4p:3q=2:5---[\text{A1}]$
2	**(a)** 1 cm : 80 000 cm 1 cm : 0.8 km Distance on map = $1.5\times0.8=1.2km$ **(b)** 1 cm^2 : 0.64 km^2 --- [M1] Area on map = $1.85\div0.64$ = 2.89 cm^2 (3sf)
3	**(a)** $T=\frac{2\pi}{\sqrt{g}}\sqrt{L}$ Since g is constant at a fixed location, $\underline{\frac{2\pi}{\sqrt{g}}}$ **<u>is a constant</u>**, so T is directly proportional to $\sqrt{L}$. [B1] **(b)** $T=k\sqrt{L}$, for k is a constant $k=\frac{T}{\sqrt{L}}$ When new L = 4L New T $=\frac{T}{\sqrt{L}}\times\sqrt{4L}$ $=2T$ % change $=\frac{2T-T}{T}\times100\%$ $=100\%---[\text{A1}]$

4	**(a)**	$t=\frac{k}{n}\Rightarrow tn=k$, where k is a constant When $t=40$, $n=7$ $\Rightarrow$ $k=7\times 40=280$ Hence, $t=\frac{280}{n}$ When $n=8$, $t=\frac{280}{8}=35$ minutes
	(b)	When $t=40-12=28$ $\Rightarrow$ $n=\frac{280}{28}=10$ It takes 10 taps to fill up the tank 12 minutes faster. Hence, 3 additional taps are needed.
5	$y=kx^2$, where k is a constant $Let\ new\ x = x_1$ $Let\ new\ y = y_1$ $x_1 = 1.25x$ $y_1 = k(1.25x)^2$ $y_1 = 1.5625kx^2$ Percentage increase in y $=\frac{y_1-y}{y}\times 100\%$ $=\frac{1.5625kx^2-kx^2}{kx^2}\times 100\%$ $=\frac{kx^2(1.5625-1)}{kx^2}\times 100\%$ $=56.25\ \%$	
6	$\frac{Area\ of\ smaller\ triangle}{Area\ of\ bigger\ triangle}=\frac{\frac{1}{2}\times B\times H1}{\frac{1}{2}\times B\times H2}$ $\frac{Area\ of\ smaller\ triangle}{Area\ of\ bigger\ triangle}=\frac{1}{3}$ $\frac{Area\ of\ Shaded\ triangle}{Diagram}$ $=\frac{1.5}{4}=\frac{3}{8}$ <u>Alternative Solution</u> Let the height of the smaller triangle be H_1. Let the height of the larger triangle be H_2. $H_2=3H_1$ Fraction of the kite that is shaded $=\frac{\frac{1}{2}\left(\frac{1}{2}\times H_2\times B\right)}{\left(\frac{1}{2}\times H_1\times B\right)+\left(\frac{1}{2}\times H_2\times B\right)}$ $=\frac{\frac{1}{2}\left(\frac{1}{2}\times 3H_1\times B\right)}{\left(\frac{1}{2}\times H_1\times B\right)+\left(\frac{1}{2}\times 3H_1\times B\right)}$ $=\frac{\frac{1}{2}\left(\frac{1}{2}\times 3H_1\times B\right)}{\left(\frac{1}{2}\times H_1\times B\right)(1+3)}$ $=\frac{3}{8}$	

7	$y=\frac{k}{\sqrt[3]{3x+2}}$ $3=\frac{k}{\sqrt[3]{3(2)+2}}$ $k=3\times\sqrt[3]{8}$ $k=6$ $y=\frac{6}{\sqrt[3]{3x+2}}$
8	$50=\frac{k}{\sqrt[3]{x}}$ $y_{new}=\frac{k}{\sqrt[3]{8x}}=\frac{k}{2\sqrt[3]{x}}$ $y_{new}=\frac{1}{2}(50)=25$
9	$A:B=\frac{1}{6}:\frac{1}{5}$ $A:B=\frac{15}{6}:\frac{15}{5}$ $A:B=\frac{5}{2}:3$ or $A:B=5:6$ $A:C=\frac{5}{2}:7$ $A:C=5:14$
10	$F=\frac{k}{r^2}$ $64=\frac{k}{r^2}$ $F=\frac{k}{(0.8r)^2}$ $F=\frac{k}{0.64r^2}$ $F=\frac{64}{0.64}$ $F=100$ units